

BRAC University
Department of Electrical and Electronic Engineering
EEE/ECE-103IL: Computer Programming Laboratory
Summer 2026 (Section: 02)

Lab Final

1. Develop a **C program** that dynamically allocates memory to store the weights of **N packages** in a warehouse. The value of **N** and the package weights will be provided by the user. The package weights are stored in the order they arrive, which does not follow any specific pattern. [CO4] (10)

For efficient loading:

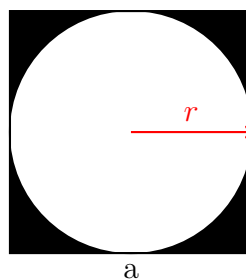
- **Lighter packages must be loaded first**
- **Heavier packages must be loaded last**

Write a C program that performs the following tasks:

- Dynamically allocate memory to store the package weights
- Read **N** package weights from the user
- Arrange the data according to the loading requirement
- Display the final loading sequence

Constraints:

- Dynamic memory allocation mandatory
 - Do not use any built-in functions
2. The diagram below shows a circle inscribed in a square with both shapes sharing the same center. [CO4] (8)



Let a be the side length of the square (user input). Write a C program to:

- Calculate the area of the square and circle
- Calculate the percentage of the total square area that lies outside the inscribed circle (the black portion) and display the result up to two decimal places

Use the formula:

$$\text{Area of circle} = \pi r^2$$

3. The following program is supposed to print **all integers** that are not divisible by 3 and 5 and lie between 1 and a user-defined number (inclusive). The program also counts the total **number** of such integers and prints the result. [CO4] (7)

No.	Code	Explanation/Corrected
1		
2	int main() {	
3	int end;	
4	printf("Enter the end number: ");	
5	scanf("%f", &end);	
6	int i, Count;	
7	count = 0;	
8	for(i=1; i<end; i++) {	
9	if ((i/3 != 0) & (i/5 != 0)) {	
10	count = count + 1;	
11	printf("%d\n", i)	
12	}	
13		
14	printf("Total count is: %c\n", Count);	
15	}	